

KAMEN-NA-OBI, Russian Federation

WMO: 298220

Lat: 53.8189N

Lon: 81.2768E

Elev: 129

StdP: 99.78

Time Zone: 7.00 (E07)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -37.5	(c) -34.0	(d) -40.1	(e) 0.1	(f) -37.3	(g) -36.6	(h) 0.1	(i) -33.7	(j) 10.6	(k) -6.3	(l) 9.8	(m) -7.5	(n) 1.0	(o) 250	(p) 0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.7	(c) 30.7	(d) 18.7	(e) 28.9	(f) 18.3	(g) 27.2	(h) 17.7	(i) 20.7	(j) 27.1	(k) 19.9	(l) 26.1	(m) 19.0	(n) 25.1	(o) 3.8	(p) 70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	13.8	23.6	17.6	12.8	22.6	16.8	12.1	21.8	60.8	27.1	57.7	26.0	54.8	25.2	
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Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.5	(b) 8.2	(c) 7.1	(e) -40.5	(f) 33.8	(g) 4.3	(h) 2.2	(i) -43.6	(j) 35.4	(k) -46.2	(l) 36.7	(m) -48.6	(n) 37.9	(o) -51.7	(p) 39.4		
(4) 9.5	(5) 8.2	(6) 7.1	(7) -40.1	(8) 22.6	(9) 4.1	(10) 1.3	(11) -43.0	(12) 23.5	(13) -45.4	(14) 24.3	(15) -47.7	(16) 25.0	(17) -50.6	(18) 26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.1	-18.8	-15.7	-7.3	4.3	12.2	18.2	19.7	17.3	10.8	3.4	-6.3	-14.2		
	DBStd	14.81	9.06	7.61	6.78	5.72	5.34	4.15	3.27	3.67	4.66	5.14	7.74	8.41		
	HDD10.0	3876	894	721	536	186	41	2	0	1	45	212	488	751		
	HDD18.3	6098	1152	954	794	421	198	52	23	63	230	463	738	1009		
	CDD10.0	977	0	0	0	16	109	249	301	227	68	7	0	0		
	CDD18.3	159	0	0	0	0	9	48	66	32	4	0	0	0		
	CDH23.3	1713	0	0	0	9	153	508	594	372	76	0	0	0		
	CDH26.7	481	0	0	0	1	34	143	176	104	22	0	0	0		
(14) Wind	WSAvg	2.8	2.8	2.8	3.0	3.3	3.4	2.6	2.1	2.2	2.4	2.9	3.2	3.3		
(15) Precipitation	PrecAvg	355	16	15	15	20	31	42	66	38	35	28	27	23		
	PrecMax	468	32	39	35	47	77	87	158	72	80	71	51	48		
	PrecMin	226	2	0	0	1	3	10	19	10	3	10	12	10		
	PrecStd	72	8	8	9	12	18	23	33	17	20	17	11	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.2	1.3	7.9	24.5	30.7	32.2	33.1	32.5	29.8	20.3	9.3	1.8		
		MCWB	0.3	0.2	4.0	13.3	17.3	18.6	20.0	19.9	17.2	12.0	6.4	0.6		
	2%	DB	-1.3	-0.7	4.8	20.2	27.3	30.1	30.5	29.4	25.2	16.8	6.8	0.4		
		MCWB	-2.3	-1.5	2.3	11.6	15.4	18.0	19.5	19.1	15.3	10.1	4.5	-0.5		
	5%	DB	-3.6	-2.4	3.0	16.8	25.0	28.2	28.6	27.2	22.2	13.9	4.5	-1.3		
		MCWB	-4.3	-3.3	1.2	9.6	14.4	17.8	19.1	17.9	14.0	8.8	2.9	-2.2		
	10%	DB	-5.6	-4.4	1.5	13.5	22.3	26.4	26.7	25.1	19.3	10.9	2.5	-3.2		
		MCWB	-6.4	-5.2	0.2	8.1	13.5	17.3	18.6	17.3	12.8	7.3	1.3	-3.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.4	0.6	4.4	14.1	18.8	21.1	22.3	21.6	18.2	12.5	7.0	0.7		
		MCDB	0.9	1.2	7.2	22.6	27.8	28.0	28.7	27.8	27.0	17.7	9.2	1.5		
	2%	WB	-2.2	-1.5	2.6	12.1	16.9	19.8	21.2	20.1	16.0	10.7	4.6	-0.4		
		MCDB	-1.4	-0.7	4.4	19.2	24.7	26.4	27.2	26.9	23.5	15.6	6.2	0.4		
	5%	WB	-4.3	-3.2	1.4	10.2	15.3	18.9	20.3	19.0	14.6	9.1	3.1	-2.1		
		MCDB	-3.6	-2.4	2.9	16.1	22.7	25.7	26.1	25.4	20.8	13.2	4.6	-1.4		
	10%	WB	-6.3	-5.1	0.2	8.2	14.1	18.1	19.5	17.9	13.2	7.6	1.3	-4.0		
		MCDB	-5.6	-4.4	1.4	13.4	21.3	24.5	25.0	23.8	18.5	10.9	2.4	-3.1		
(35) Mean Daily Temperature Range	MDBR		9.3	10.2	10.1	10.6	12.4	11.5	10.7	12.2	11.8	9.0	6.7	8.2		
	5% DB	MCDBR	8.6	8.4	9.1	16.0	16.5	14.7	13.8	15.7	17.2	14.7	7.0	8.3		
		MCWBR	8.5	8.1	7.2	8.6	7.4	5.9	5.2	6.7	8.1	8.6	5.5	8.1		
	5% WB	MCDBR	8.7	8.5	8.1	15.3	14.2	12.0	11.3	13.2	14.6	13.4	6.8	8.1		
		MCWBR	8.6	8.3	6.6	8.7	7.3	5.6	5.2	5.9	7.6	8.3	5.5	8.0		
(40) Clear-Sky Solar Irradiance	taub		0.298	0.331	0.384	0.411	0.413	0.411	0.437	0.416	0.374	0.356	0.335	0.302		
	taud		2.013	1.957	1.902	2.114	2.191	2.283	2.210	2.243	2.306	2.234	2.064	1.965		
	Ebn at Noon		614	720	764	811	834	839	806	800	789	703	560	508		
	Edh at Noon		76	112	149	138	135	125	132	121	100	85	72	65		
(44) All-Sky Solar Radiation	RadAvg		1.07	2.13	3.60	4.73	5.67	6.17	5.55	4.78	3.35	1.75	0.85	0.72		
	RadStd		0.09	0.13	0.29	0.30	0.54	0.49	0.36	0.34	0.34	0.23	0.14	0.10		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page